

Ensemble Learning

BY

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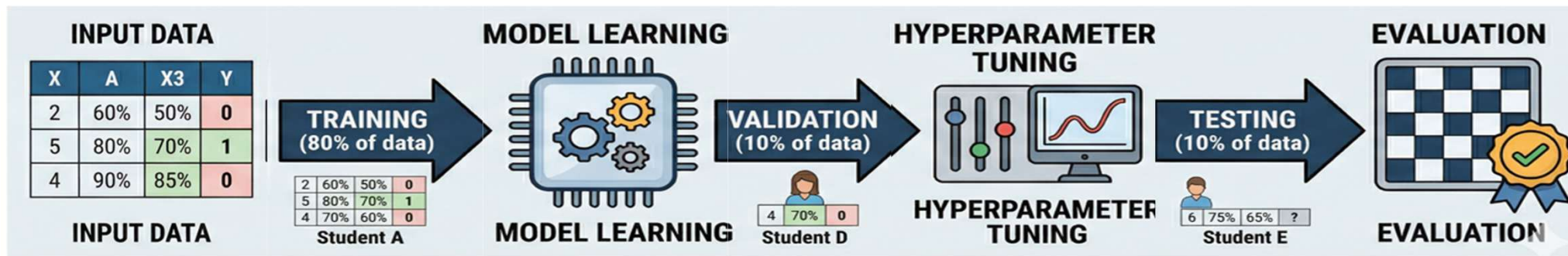
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Traditional Approach

	 STUDY HOURS (X1)	 ATTENDANCE (%) (X2)	 ASSIGNMENT SCORE (X3)	 RESULT (Y) (0: Fail / 1: Pass)
Student A	2	60%	50%	 0 (Fail)
Student B	5	80%	70%	 1 (Pass)
Student C	7 hours	90%	85%	 1 (Pass)
Student D	4 hours	70%	60%	 0 (Fail)

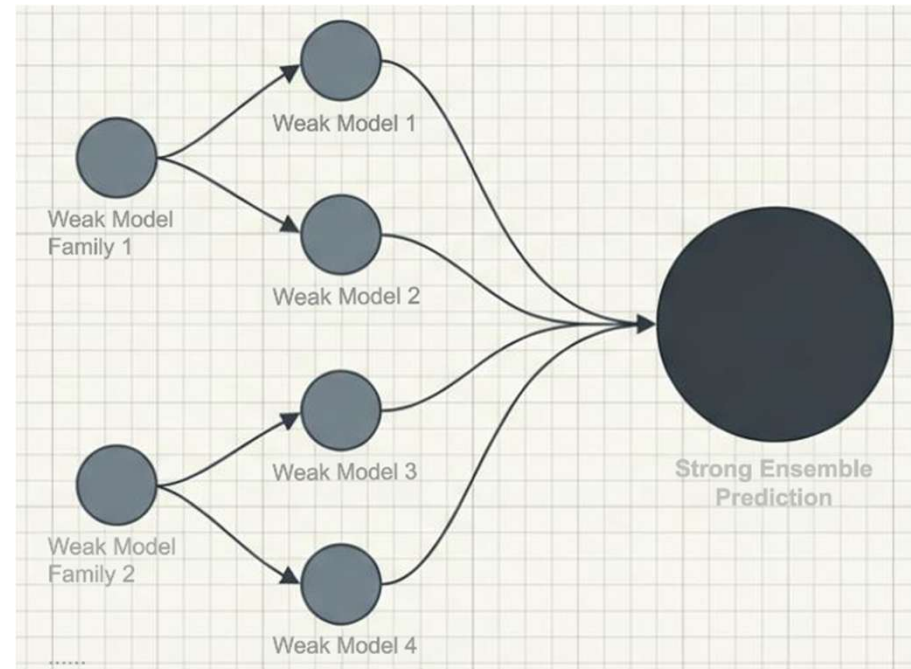
Traditional Approach



Ensemble Learning

- **Definition:**

- Ensemble learning is a powerful machine learning technique where multiple models (**called base learners or weak learners**) are combined to improve prediction accuracy.
- Instead of relying on a single model, ensemble methods leverage the strength of multiple models to reduce errors.

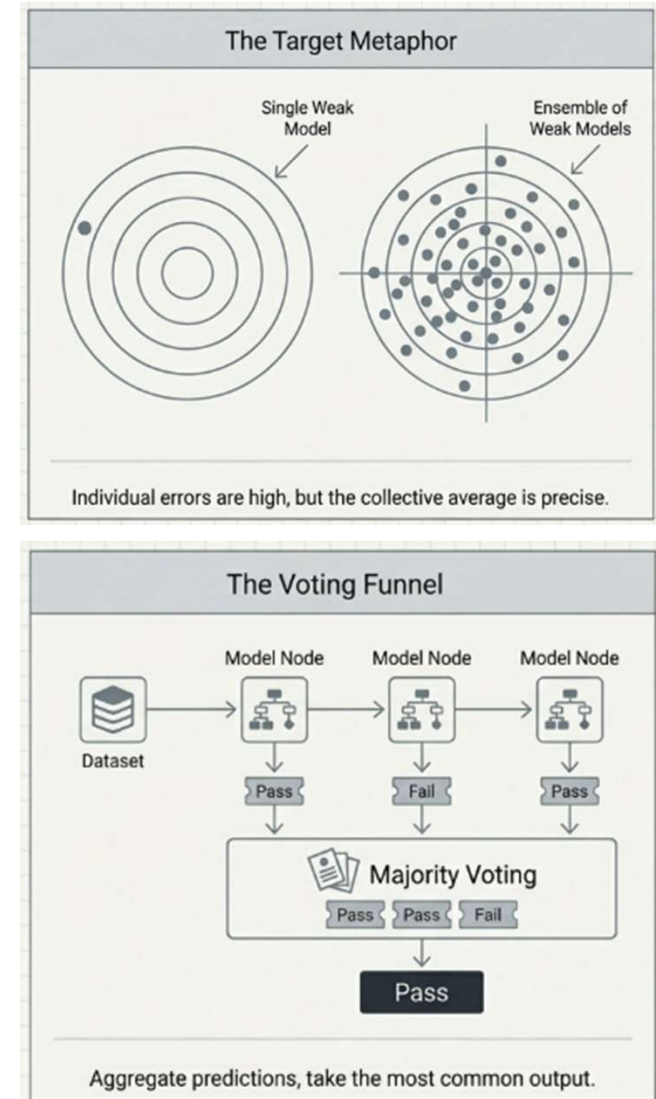


Intuition: Ensemble Learning

- Many weak learners acting together form a single strong learner. Errors cancel each other out.

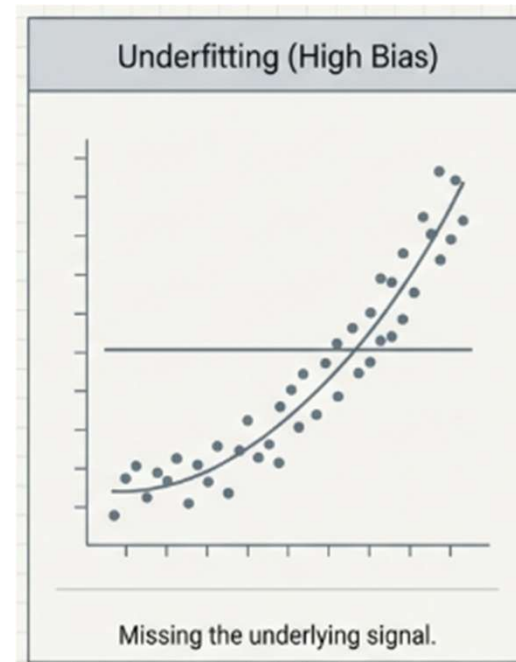
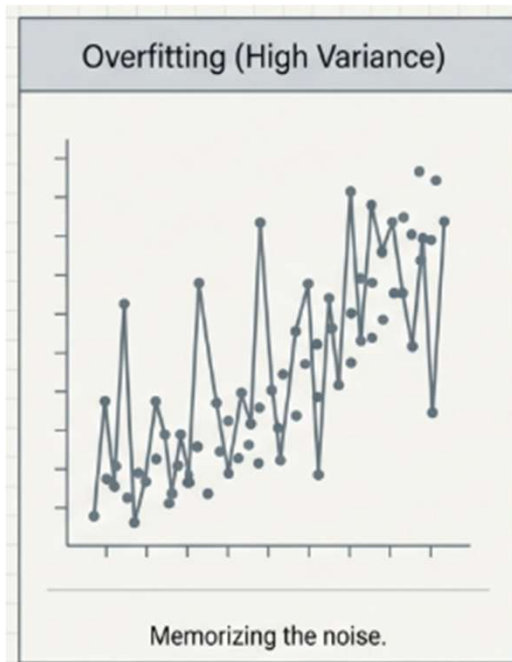
Example:

- One student guesses exam answers → poor performance
- 50 students vote → better accuracy
- This is the wisdom of the crowd.



Why use ensemble learning ?

- Single models often suffer from:

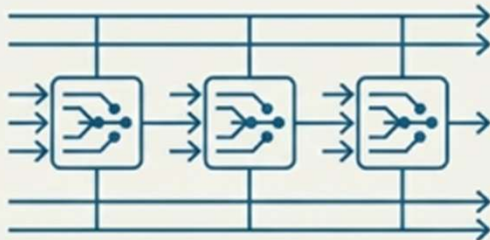


Types of Ensemble Learning

- Bagging:
 - bagging reduces variance by training models independently
- Boosting:
 - boosting reduces bias by training models sequentially
- Stacking:
 - stacking combines different models using a meta-learner.
- Examples:
 - Popular algorithms like Random Forest and XGBoost are widely used in real-world applications such as healthcare, banking, and education.

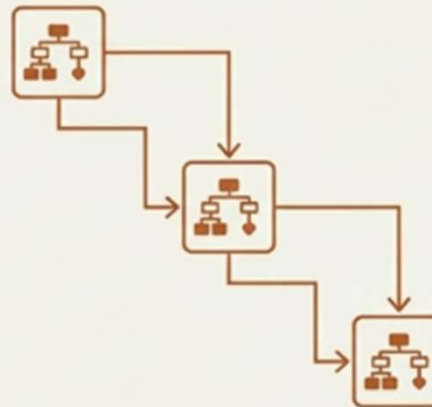
Architectural Overview: The Ensemble Learning

Bagging



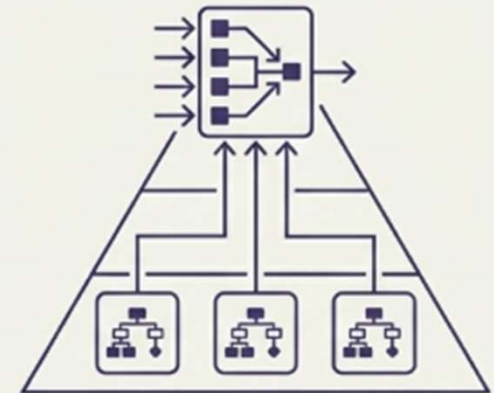
Parallel training on random data subsets to reduce variance.

Boosting



Sequential training focused on previous errors to reduce bias.

Stacking



Hierarchical training using a meta-learner to optimize combinations.

1. Bagging (Bootstrap Aggregating)

- **Idea:**

- Train multiple models independently using random subsets of data
- Combine predictions (usually averaging or voting)

- **Steps:**

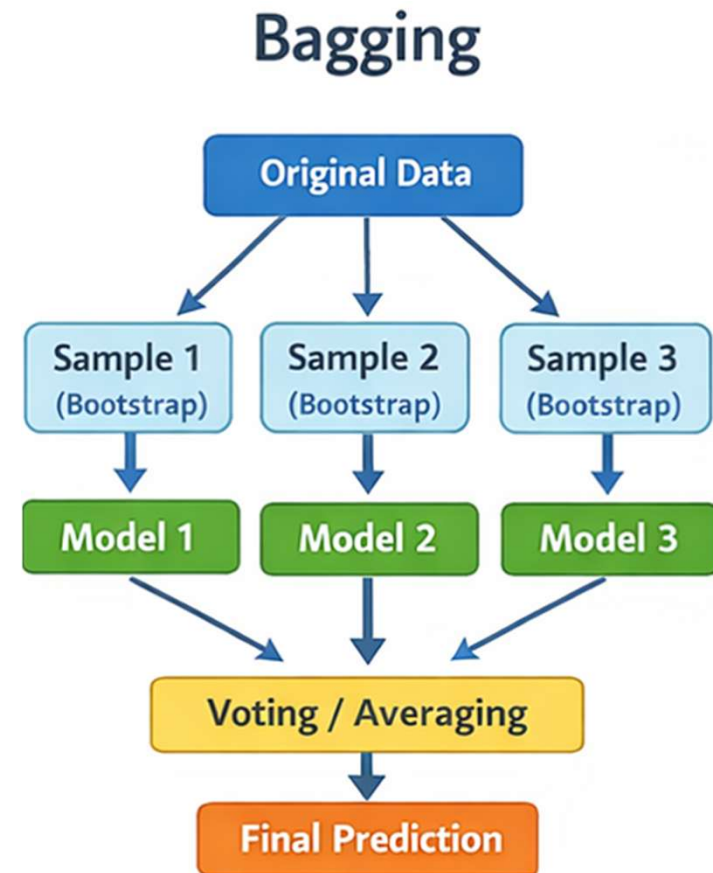
- Create multiple random datasets (with replacement)
- Train a model on each dataset
- Aggregate predictions

- **Key Feature:**

- Reduces **variance**

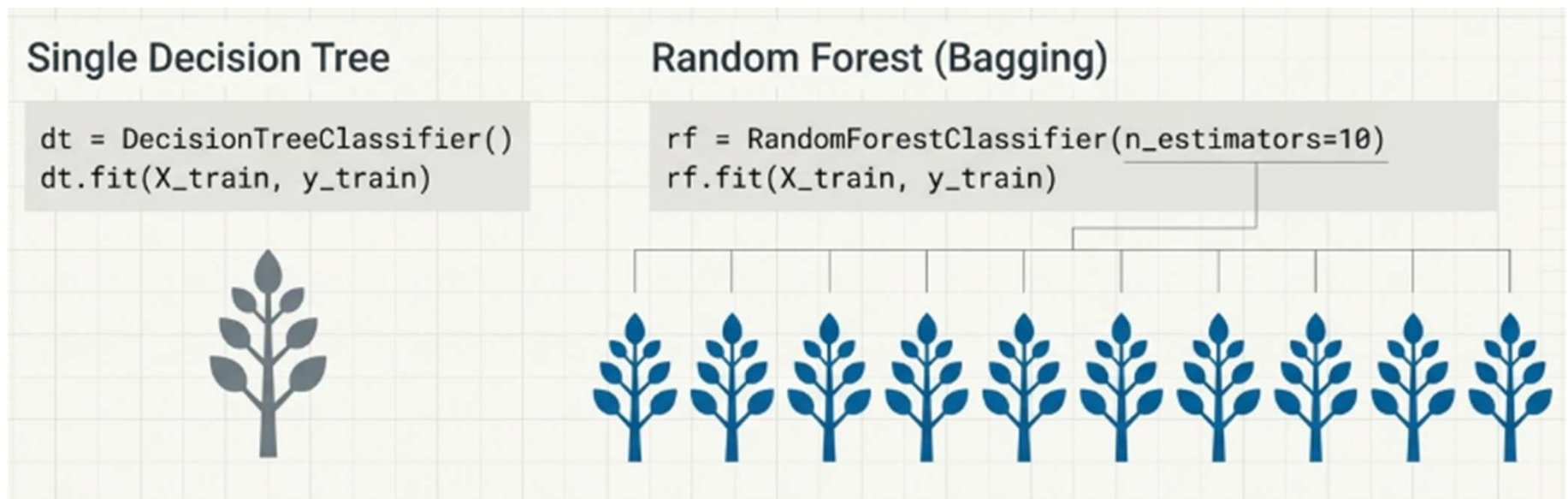
- **Example Algorithm:**

- Random Forest



Single Model vs Bagging

- With a single parameter changes, we upgrade from an unstable alone tree to an entire voting forest.



2. Boosting

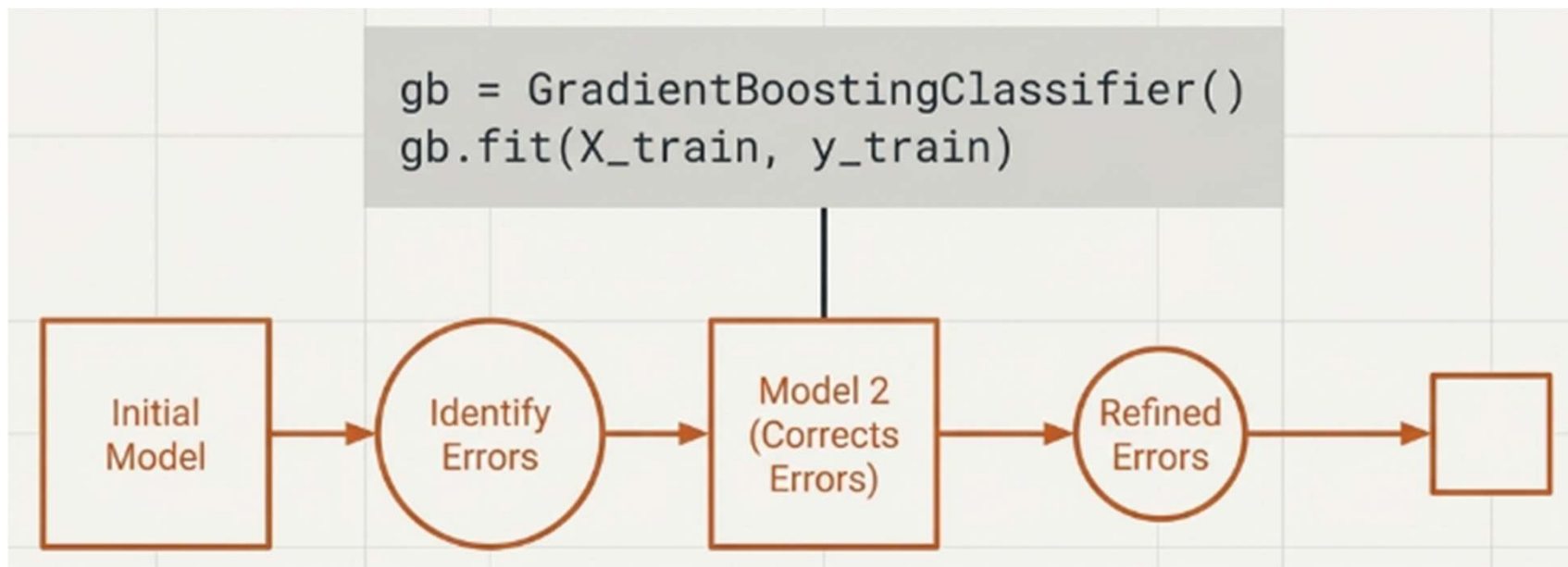
- **Idea:**
 - Train models sequentially
 - Each new model focuses on errors made by previous models
- **Steps:**
 - Train first model
 - Increase weight of misclassified data
 - Train next model on harder cases
 - Repeat
- **Key Feature:**
 - Reduces bias
- **Popular Algorithms:**
 - AdaBoost
 - Gradient Boosting
 - XGBoost

Boosting



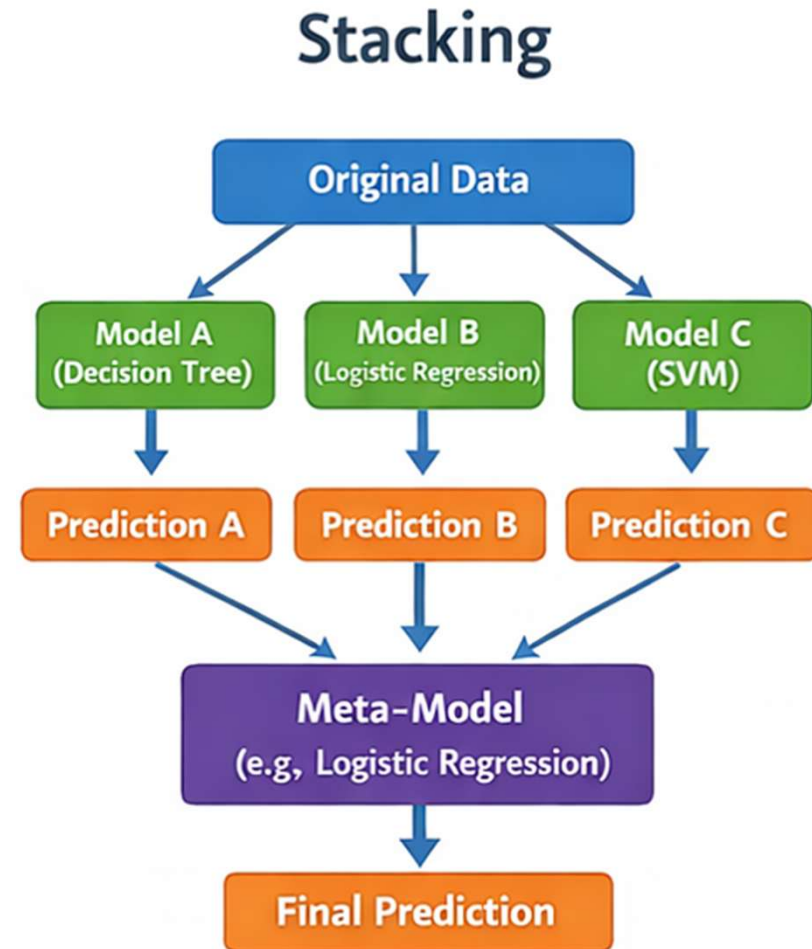
Sequential Boosting

- Instead of training parallel voters, GradientBostingClassifier constructs a disciplined sequential pipeline where every new model hunts down the exact errors missed by the previous model.



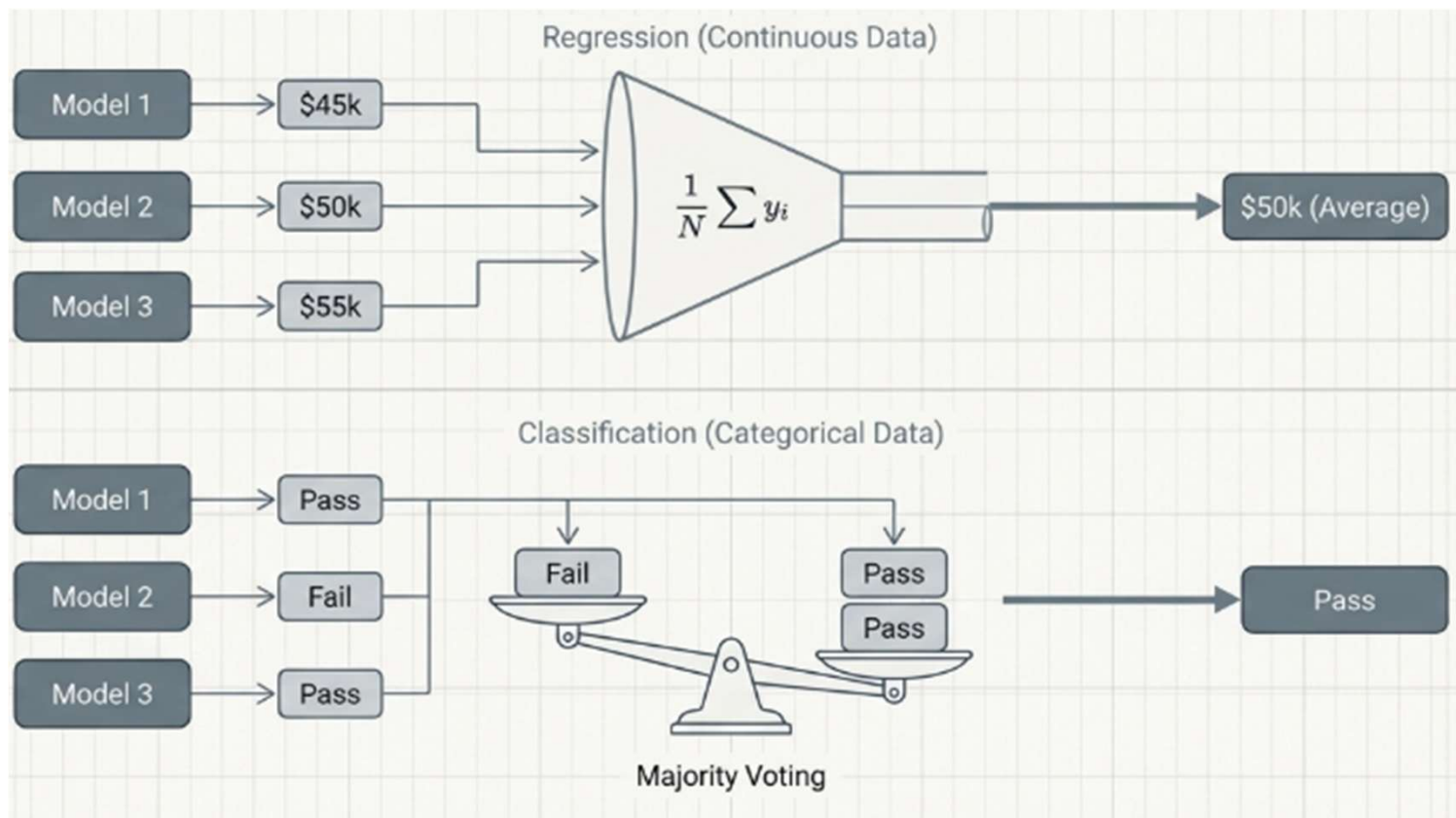
3. Stacking (Stacked Generalization)

- **Idea:**
 - Combine different models using a meta-model
- **Steps:**
 - Train multiple base models
 - Use their predictions as input
 - Train a final model (meta-learner)
- **Key Feature:**
 - Learns how to combine models optimally



Mathematical Intuition

The aggregation mechanism adapts to the data type. Averages smooth out numerical variance, while majority voting democratizes categorical decisions.



Bias-Variance Tradeoff

Method	Bias	Variance
Bagging	Same	Reduced
Boosting	Reduced	Slightly increased
Stacking	Reduced	Reduced

The Accuracy Jump & Final Tradeoffs

- Ensemble learning reliably delivers superior predictive accuracy, but it demands a tradeoff in computational cost and interpretability.



Key Takeaways

- Ensemble = combination of models
- Bagging → reduces variance
- Boosting → reduces bias
- Stacking → smart combination
- Used in real-world high-performance systems